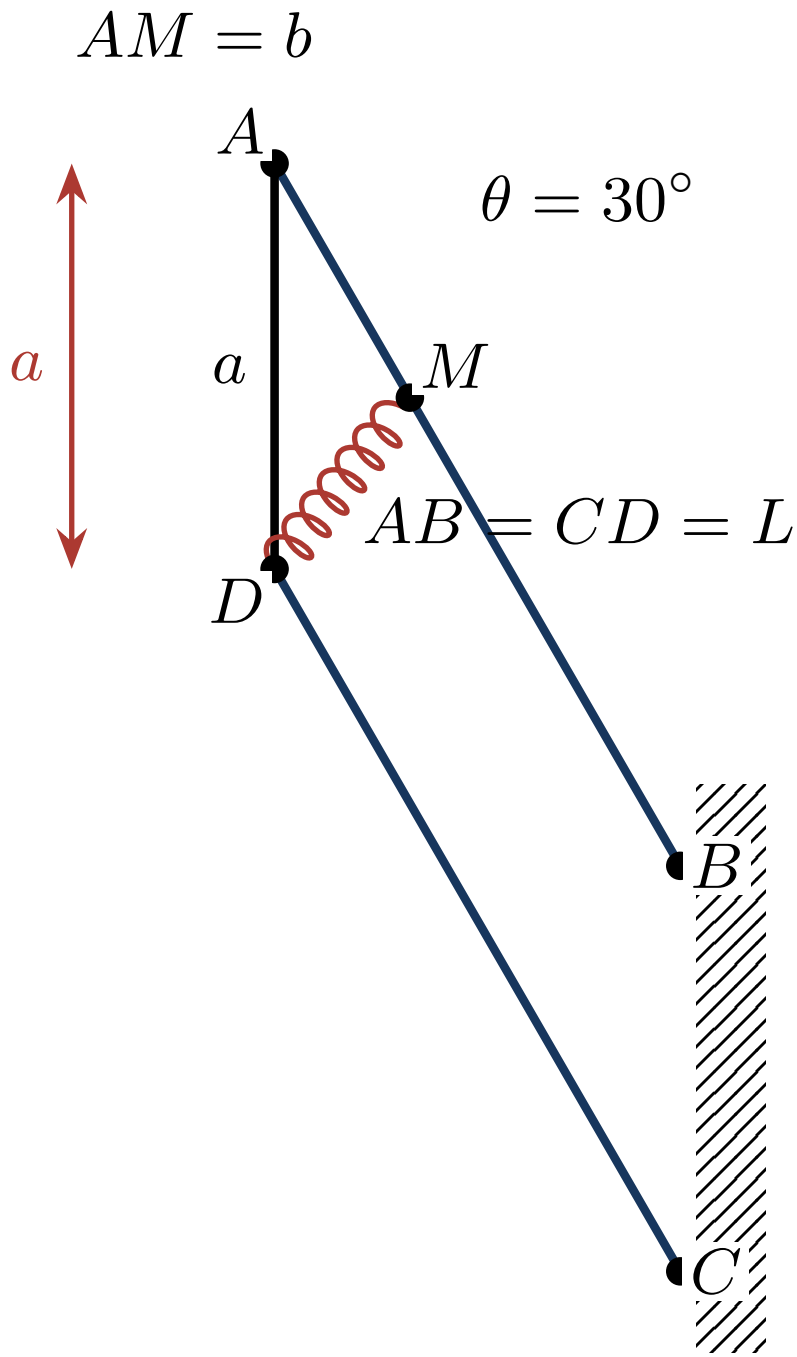

$$MD^2 = a^2 + b^2 - 2ab\cos\theta$$

$$y_A = a + L\cos\theta, \quad y_{CD} = L\cos\theta/2$$